Distance vs. displacement

02 H

DIRECTIONS · Fill all four boxes for each term. Use a number-line sketch in at least one box.

Distance

DEFINITION · MY WORDS

HOW TO FIND IT

Σ |each leg| = ____

EXAMPLES

NON-EXAMPLES

Displacement

DEFINITION · MY WORDS

FORMULA + SIGN

Δx = ____ − ____

SKETCH AN EXAMPLE

HOW IT DIFFERS FROM DISTANCE

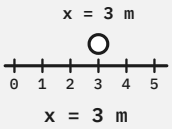
One sentence using both terms: _____

K1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: where and how far

Language goal: I can **compare** distance and displacement.



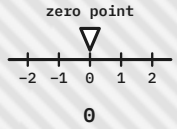
position x

posición · posição

where it is, from zero

my language: _____

use it: _____



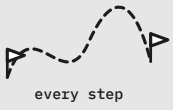
origin

origen · origem

the zero point

my language: _____

use it: _____



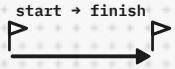
distance

distancia · distância

the whole path length

my language: _____

use it: _____



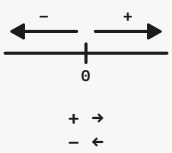
displacement

desplazamiento · deslocamento

change in position, with direction

my language: _____

use it: _____



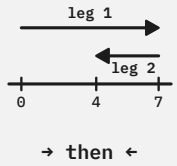
direction

dirección · direção

which way: + or −

my language: _____

use it: _____



leg

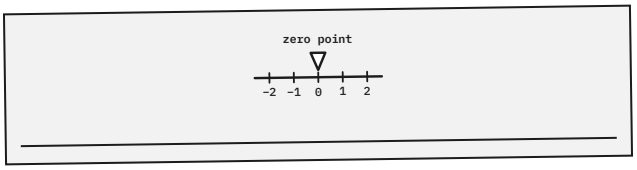
tramo · trecho

one part of a trip

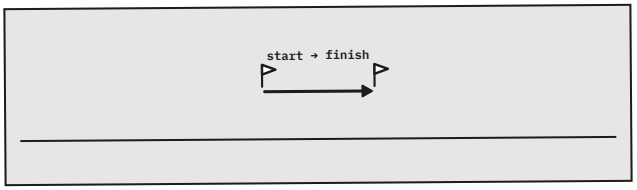
my language: _____

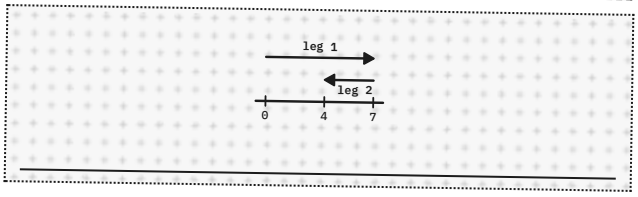
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

K1 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Fix the group chat

02 R

DIRECTIONS · Reply in the empty bubbles. Show the math on the right.

Hallway walk crew

ELI

walked 4 m to the door and back. my displacement = 8 m

YOU ①

MAZ

my locker is at x = −3 m. negative means I moved left, right?

YOU ②

NADIA

I went −3 m → +2 m → −1 m. distance?

YOU ③

ELI

and her displacement??

YOU ④

SHOW YOUR RECEIPTS

① distance vs. displacement

② position vs. motion

③ Σ |each leg|

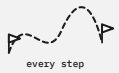
④ Δx = x_f − x_i

K1 1 2 3

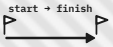
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it

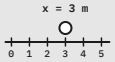
Language goal: I can **compare** distance and displacement.



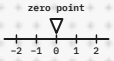
distance
distancia · distância
all of the path



displacement
desplazamiento · deslocamento
start → finish



position
posición · posição
where it is



origin
origen · origem
the zero point

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: distance · displacement · position · zero · direction

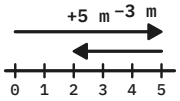
My ____ tells where I am from a ____ point. ____ counts every step I walk. ____ compares only start and finish, and it has a ____.

2 · SAY IT TO A PARTNER, THEN WRITE IT



I walked ____ m in total, so my distance is ____ m.
I ended ____ m from my start, so my displacement is ____ m.

3 · SOLVE WITH THE STEPS



Walk from 0 m to 5 m, then back to 2 m.

STEP 1 · ADD THE LEGS
____ + ____ = ____ m

STEP 2 · END − START
____ − ____ = ____ m

STEP 3 · SAY IT
"Distance ____ m, displacement ____ m."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ Distance and displacement are different because ____ .
☐ My displacement is zero because ____ .
☐ First I was at ____, then ____, so ____ .

I said it ☐ · I wrote it ☐

K1 1 2 3

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UNIT 1 · DAY 2 · K1 · CASE FILE

02 T

The missing hall pass

Hallway cameras logged a student's position (door = 0 m, east = +). Reconstruct the route.

EXHIBIT A · CAMERA LOG

TIME	POSITION	LOCATION
8:01	0 m	door
8:02	+12 m	water fountain
8:04	−5 m	locker
8:05	+3 m	classroom

EXHIBIT B · DRAW THE ROUTE ON A NUMBER LINE (−6 to +14 m)

ANALYSIS 1 · DISTANCE

Add every leg: |12 − 0| + |−5 − 12| + |3 − (−5)|

ANALYSIS 2 · DISPLACEMENT

Δx = final − start

WITNESS STATEMENT · TRUE OR FALSE?

"She only went 3 m — that's her displacement, so that's how far she walked."

THE VERDICT

Distance ____ m · displacement ____ m. They differ because ____

K1 1 2 3

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☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Position, distance, or displacement?

02 N

DIRECTIONS · Write each card's letter in its column.

A "3 m east of the door"	**B** 400 m around a track	**C** "ended 2 m left of start"	**D** never negative
E $\Delta x = x_f - x_i$	**F** $x = -2$ m	**G** counts every leg	**H** can be 0 after a long trip
I needs an origin	**J** a scalar	**K** a vector	**L** a signed coordinate

POSITION	DISTANCE	DISPLACEMENT
where it is	total path	start → finish

EXPLAIN ONE TRICKY CARD

Card ____ belongs where it does because _____

K1 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else